LINEAR REGRESSION MODEL REPORT

Insurance Charges Prediction Analysis

PoE Task 1

INTRODUCTION

This report presents the steps followed to prepare, explore, and model the insurance dataset using Linear Regression. The objective of this study was to build a predictive model capable of estimating insurance charges based on customer attributes such as age, BMI, smoking status, and region.

REPORT OUTLINE

- Data cleaning procedures
- Exploratory Data Analysis (EDA)
- Feature selection process
- Model training
- Model evaluation
- Model improvement through retraining

These steps ensured the dataset was suitable for modelling and that the final model produced reliable predictions.

DATASET OVERVIEW

- The dataset used in this study contains information about insurance customers and their associated medical costs.

Feature	Description
Age	Age of the individual
Sex	Gender of the individual
Bmi	Body Mass Index
Children	Number of dependents
Smoker	Smoking status
Region	Residential region
Charges	Insurance cost (target Variable)

DATASET OVERVIEW CONT…

- **Target Variable**
  - **Charges**
    - This variable represents the medical insurance costs and is the value predicted using Linear Regression.

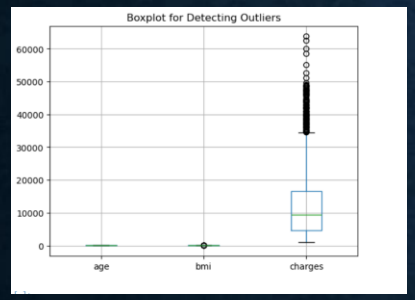
DATA CLEANING PROCESS

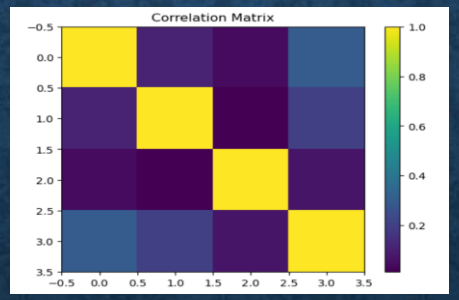
- Before training the model, the dataset was cleaned to ensure accuracy and reliability.

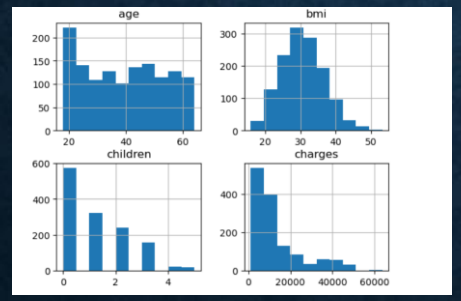
Steps Taken	Results
Checking Missing Values	The dataset was checked for missing or null values. No missing values were found in the dataset
Checking Duplicate Records	Duplicate records were removed. This ensured data consistency
Encoding Categorial Variables	Categorical variables such as sex, smoker, region were converted into numerical values. This step allowed the model to process categorical data
Feature Scaling	Ensured all variables contributed equally to model training.

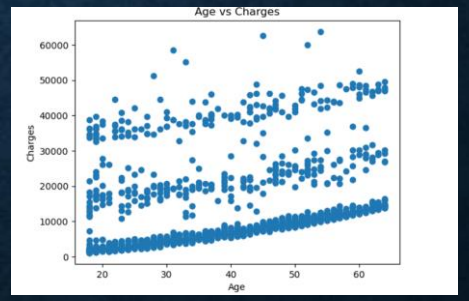
EXPLORATORY DATA ANALYSIS (EDA)

- EDA was conducted to understand the structure and relationships within the dataset.
- Visualizations were used to identify patterns, distributions, and correlations.









DISTRIBUTION OF NUMERICAL VARIABLES

- Histograms were created to visualize:
- age
- bmi
- charges

Key Findings:

- Charges were positively skewed
- Most individuals had BMI between 25–35
- Age distribution showed a balanced spread

RELATIONSHIP BETWEEN SMOKING AND CHARGES

- A boxplot was used to compare smoker vs non-smoker costs.

Key Findings:

- Smokers had significantly higher insurance charges
- Smoking status was identified as a strong predictor
- This variable was considered highly important for modelling.

CORRELATION ANALYSIS

- A correlation heatmap was created to identify relationships between variables.

Key Findings:

- Smoking status had the strongest correlation with charges
- Age showed moderate correlation
- BMI showed mild correlation
- This confirmed that these features were relevant for prediction.

FEATURE SELECTION

- Feature selection was performed to ensure only relevant predictors were used.

Method Used:

- Backward elimination and correlation analysis.

Selected Features:

- age
- bmi
- smoker
- children
- region
- sex
- Features with low statistical importance were removed when necessary.
- This improved model efficiency and reduced noise.

MODEL TRAINING

- A Linear Regression model was trained using the prepared dataset.

SPLITTING DATASET

Splitting Dataset

- The dataset was divided into:
- Training data (80%)
- Testing data (20%)

This ensured unbiased evaluation.

INITIAL MODEL TRAINING

- A Linear Regression model was trained. The model learned relationships between predictors and insurance charges.

MODEL EVALUATION

- Model performance was evaluated using multiple metrics.
- Using multiple metrics provides a complete understanding of model accuracy.

R-SQUARED (R^2)

- Measures how much variation in charges is explained by the model.

Interpretation:

- Higher values indicate better performance.

MEAN ABSOLUTE ERROR (MAE)

- Measures average prediction error.

Interpretation:

- Lower values indicate better accuracy.

MEAN SQUARED ERROR (MSE)

- Measures squared prediction errors.

Interpretation:

- Penalizes large errors.

ROOT MEAN SQUARED ERROR (RMSE)

- Measures standard prediction error magnitude.

Interpretation:

- Lower RMSE indicates better predictions.

MODEL RETRAINING AND IMPROVEMENT

- After evaluating the initial model, improvements were implemented.

Improvements made	Description
Feature scaling	This improved model performance
Ridge Regression Implementation	Regularization improved prediction stability
Result of Retraining	The retrained model showed improved: R^2 score Prediction accuracy Error reduction This confirmed that retraining enhanced model performance.

MODEL INTERPRETATION

- Model predictions were compared with actual values.
- Residual analysis was performed to examine prediction errors.

Key Observations:

- Predictions closely followed actual values
- Errors were randomly distributed
- No strong bias patterns were detected
- This confirmed the reliability of the model.

BUSINESS INSIGHTS FROM THE MODEL

- The analysis revealed several important insights:
- Smoking significantly increases insurance charges
- Higher BMI slightly increases costs
- Age contributes to higher medical expenses
- Customers with more dependents may incur higher costs

These insights can help insurers:

- Assess customer risk
- Design pricing strategies
- Improve cost predictions

MODEL LIMITATIONS

- Although the model performed well, some limitations exist:
- Linear Regression assumes linear relationships
- External factors affecting charges were not included
- Outliers may affect prediction accuracy

Future improvements may include:

- Using more advanced models
- Adding additional predictive features

CONCLUSION

- This project successfully developed a Linear Regression model to predict insurance charges.

The following steps were completed:

- Data cleaning and preprocessing
- Exploratory Data Analysis
- Feature selection
- Model training
- Model evaluation
- Model retraining and improvement

The final model demonstrated reliable predictive performance and provided valuable insights into the factors affecting insurance charges.

This model can assist organizations in improving decision-making and cost prediction processes.

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